

RESEARCH MANAGEMENT SYSTEM- UPGRADE

Introduction

This document defines a comprehensive research management platform for Nyahururu Hospital. The system manages research projects from proposal submission through publication.

Objectives

Provide proposal review, automated M-Pesa payment processing, reviewer workflows, ethics approval, progress monitoring, final paper review, publication management, certificate generation, and audit tracking.

User Roles

Researcher — Registers, submits research proposals, progress reports, and final papers. Tracks submission status and downloads own certificates and revenue reports.

Reviewer — Invited and managed by the Admin. Evaluates assigned proposals, progress reports, and final papers, scoring originality, relevance, feasibility, ethics compliance, and expected impact. Submits decisions (Approved, Revision Needed, Rejected, or Suspend Study for progress-stage submissions) with comments.

Research Committee — A reviewer may be promoted to the Research Committee, or a committee member may be invited directly, by the Admin. Once a reviewer has approved a final paper, the Research Committee reviews and grants final approval before publication. The Committee has visibility into all research papers across the system, regardless of reviewer assignment, to support this oversight role.

Admin — A hospital system administrator (managed through the hospital's user directory, alongside other hospital staff accounts, separate from the Researcher/Reviewer/Committee records). Invites and manages Reviewers and Research Committee members, assigns reviewers to submissions, manages download pricing, and publishes approved final papers after Research Committee sign-off. Research data and reporting are accessible from both the Research Portal dashboard and the hospital's administrative dashboard.

Research Lifecycle

Draft Proposal → M-Pesa Payment → Proposal Review → Approval & Clearance Certificate → Progress Submission → Progress Review → Final Paper Submission → Final Review → Research Committee Approval → Publication Approval → Completion Certificate → Archive.

Stage 1 - Research Proposal Submission

Researchers submit: Research Title, Abstract, Problem Statement, Background, Objectives, Hypotheses, Literature Review Summary, Expected Outcomes, Study Duration, Study Sites, Co-Investigators, Funding Source, Ethics Information, and Proposal Documents. Uploads: Proposal PDF (Should cover everything about the proposal entails, budget, timeline, ethics forms, and supporting documents).

Validation: Required fields, file type validation, duplicate title detection, plagiarism checks.

Statuses: Draft, Awaiting Payment, Submitted, Under Review, Revision Requested, Approved, Rejected.

Proposal Review Workflow

Admin assigns reviewers. Reviewers score originality, relevance, feasibility, ethics compliance, proposal PDF document, and expected impact.

Review outcomes: Approved, Revision needed, Rejected. Complete review history must be stored. Should have a reviewer comment text area.

One-Time M-Pesa Payment

Payment occurs only once during proposal submission. Uses STK Push. Stores CheckoutRequestID, MerchantRequestID, Receipt Number, Phone Number, Amount, Status, and Callback Data. Payment unlocks all subsequent stages without additional fees.

Research Approval & Clearance Certificate

Generated automatically after proposal approval. Includes Certificate Number, Research Code, Researcher Information, Institution, Study Sites, Validity Period, Committee Approval Statement, QR Verification Code, Signature Area, and Official Seal.

Stage 2 - Research Progress Submission

Purpose: Evaluate ongoing research before final publication.

Required Information: Methodology, Study Design, Sampling Method, Sample Size Achieved, Data Collection Progress, Statistical Methods, Analysis Tools, Preliminary Findings, Deviations from Approved Protocol, Ethical Incidents, Participant Withdrawal Information.

Uploads: Draft Manuscript, Datasets, Statistical Outputs, Survey Tools, Interview Guides, Supporting Documents.

Progress Review Process

Reviewers assess: Methodology Compliance, Data Quality, Statistical Validity, Ethical Compliance, Research Progress. Recommendations: Approved, Revision needed, Suspend Study.

Should have a text area for reviewers' comments.

Stage 3 - Final Paper Submission

Researchers submit: Final Manuscript, Publication Version, Final Dataset, Data Dictionary, Statistical Scripts, Consent Documentation, Ethics Approval Documents, Conflict of Interest Declarations, Funding Disclosures, AI Usage Declaration, Plagiarism Report.

Final Review Process

Reviewers validate scientific integrity, publication readiness, completeness of supporting documents, and compliance with institutional standards. The reviewer approves, requires a revision, or rejects the final paper.

Once a reviewer approves a final paper, it is forwarded to the Research Committee, which grants final approval, requests revision, or rejects the paper before it can be published.

Publication & Repository Module

Store approved research papers. Publication Date, Journal Information, Download Statistics, Repository Search, Public Research Profiles.

Issued after final approval and publication. Contains Research Information, Publication Information, Completion Statement, Director Signature Area, Verification QR Code, and Certificate Number.

Notifications

Email and in-system notifications for: Payment Confirmation, Reviewer Assignment, Revision Requests, Approval Notices, Certificate Generation, Publication Approval.

Audit Trail

Track every action: Submissions, Reviews, Status Changes, Payments, Certificate Generation, Downloads, Administrative Actions.